

Exercise Sheet 2

Deadline for submission: Wednesday, 8 October 2025 at 08:00.

Discussed in exercise session: Friday, 10 October 2025 at 10:15.

Exercise 2.1 - Solving a Linear System with LU Decomposition

For this exercise, you have to complete the code provided in the jupyter notebook `LU_Crout_empty.ipynb`.

(a) **LU Decomposition with Crout's Method**

We will decompose the matrix A into the product $L \cdot U$. To do this, we will use Crout's method. The output will then be the lower and upper triangular matrices L and U . In Python, implement this algorithm and apply it onto the following matrix A :

$$A = \begin{pmatrix} 1 & 0 & -5 \\ -6 & 2 & 9 \\ 2 & -3 & 2 \end{pmatrix}$$

* Optional: Is your implementation efficient? Think about how to speed up the computation.

(b) **Back- and Forward Substitution: solving $LUx = b$**

The decomposition $A = LU$ from above is used to solve a linear system $Ax = b$. In particular, this is a two-step process which requires forward and backward substitution. Write a Python script that uses back- and forward substitution and the LU decomposition of the matrix A calculated in the previous exercise to solve $Ax = LUx = b$ with

$$b = \begin{pmatrix} -23 \\ 54 \\ 2 \end{pmatrix}$$

What is the advantage of using an LU decomposition over Gauss-Jordan elimination?

Exercise 2.2 - Cholesky Decomposition

For symmetric positive-definite matrices, we can apply the Cholesky decomposition. In our case, matrix

$$A = \begin{pmatrix} 16 & -4 & -12 \\ -4 & 5 & 5 \\ -12 & 5 & 19 \end{pmatrix}$$

shall be written as LL^t , where

$$L = \begin{pmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{pmatrix}.$$

62060-01: Übung: Scientific Computing

It should be clear that the Cholesky decomposition is a special case of the LU decomposition which is computational *less* demanding (why?) than LU decomposition. In Python, implement the Cholesky decomposition either by modifying the code you wrote in exercise 3.1.2 or by implementing it from scratch and calculate the matrix L . Use the provided jupyter notebook `Cholesky_empty.ipynb` for this exercise.

Exercise 2.3 - Determinant**(a) Determinant of a general 2×2 Matrix**

Deduce a formula for the calculation of the determinant of a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with $a, b, c, d \in \mathbb{R}$, by using again the rules 1 – 9 for determination of the determinant from the lecture.

Note: In this exercise we do not want to use the LU-Decomposition of A to calculate the determinant, but the rules 1 – 9 from the lecture slides in chapter 1.

(b) Computation of a Determinant

Compute the determinant (by using the rules 1 – 9 for determination of the determinant from the lecture) as well as the eigenvalues by hand of the matrix

$$A = \begin{pmatrix} 3 & 2 & 2 \\ 1 & 2 & -1 \\ 2 & 2 & 3 \end{pmatrix}$$

and show that the determinant of this matrix A is the product of its eigenvalues.